Financial Econometrics I

High-frequency financial models - Realized Measures (Lecture 8)

Summer Semester 2023/2024 by Jozef Baruník

Introduction

Key problem in financial econometrics is modeling, estimation and forecasting of conditional return volatility and correlation

implications for derivatives pricing, risk management, asset allocation

conditional volatility is highly persistent

conditional volatility is unobservable if we consider traditional latent variable models like (G)ARCH, Stochastic Volatility

Moreover, all these parametric models are:
- difficult to estimate
- do not utilize high-frequency data
- standardized returns from these models are not Gaussian
- as an implication, forecasts are not precise
- multivariate extensions are difficult

New approach uses estimates of latent volatility based on high-frequency data, which is not relying on any parametric assumptions, is called *realized variance measures*
- volatility is observable
- traditional time series models are also applicable
- high dimensional multivariate modeling is feasible

Realized Volatility (Realized Variance) Construction

Let's consider $p_{i,t}$, log-price of asset i at time t

n is fraction of a trading session associated with sampling frequency

$m = 1/n$, number of sampled observations per trading session

T is number of days in the sampe (mT is number of total observations)

Example on FX market

Consider we would have EUR futures prices sampled every 5 minutes in 24/7 trading setting

m = 288, n=1/288

Thus we have 288 observations per day

Realized Variance

Realized Variance for asset *i* on day *t* can be constructed as:

$\mathrm{RV}_{i,t} = \sum_{j=1}^{m} r_{i,t-1+j\,n}^2$
for t=1,...,Tdays.

Realized Volatility

Realized Volatility for asset *i* on day *t* can be constructed as:

$\mathrm{RVOL}_{i,t} = \sqrt{\mathrm{RV}_{i,t}}$
for t=1,...,Tdays.

Realized log-volatility

Realized log-Volatility for asset *i* on day *t* can be constructed as:

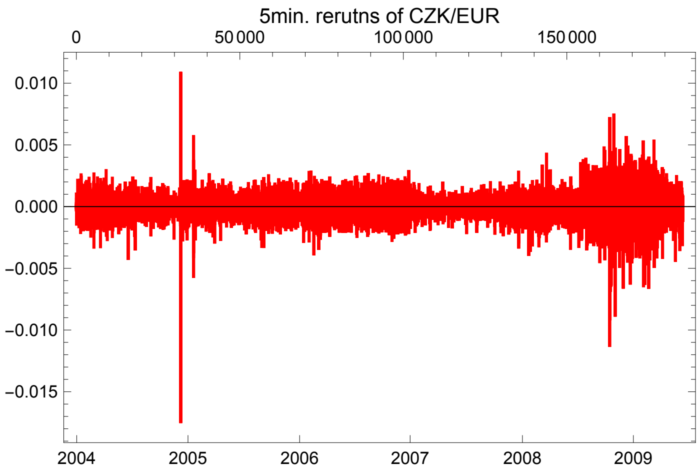
$\mathrm{RVOL}_{i,t} = \ln(\mathrm{RVOL}_{i,t})$
for t=1,...,Tdays.

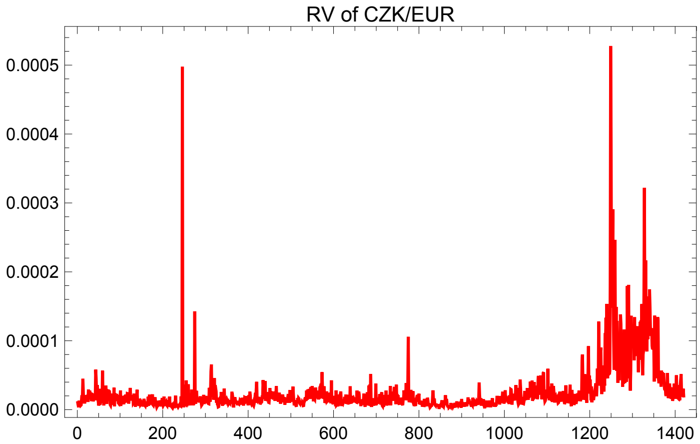
Example of CZK data

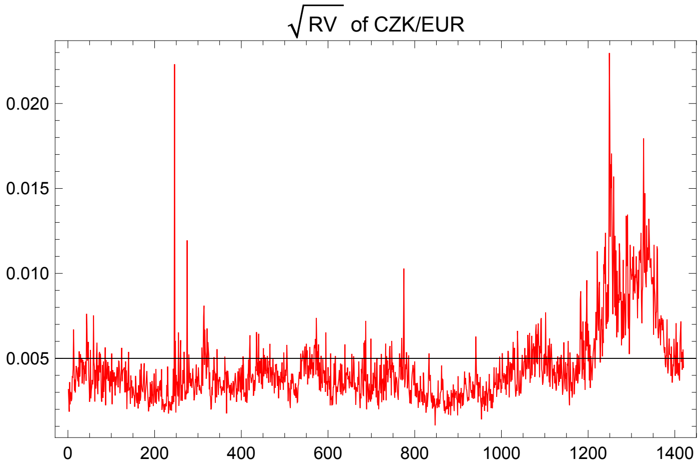
Let’s compute Realized Variance, Volatility and Log-volatility of CZK/EUR series for the period of 2004-6/2009 on 5-minute data.

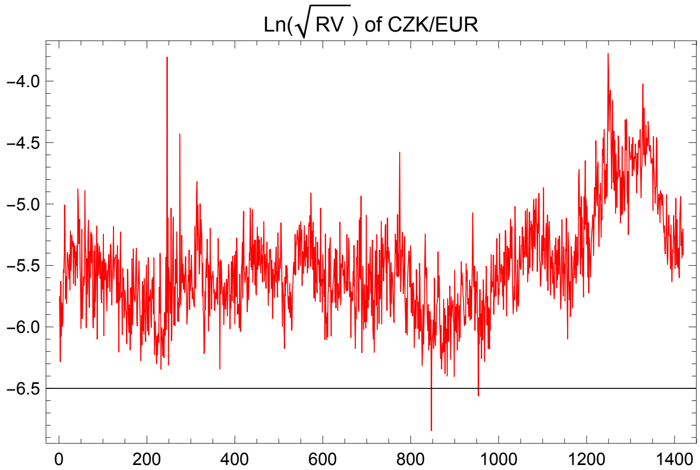
Code

Plots

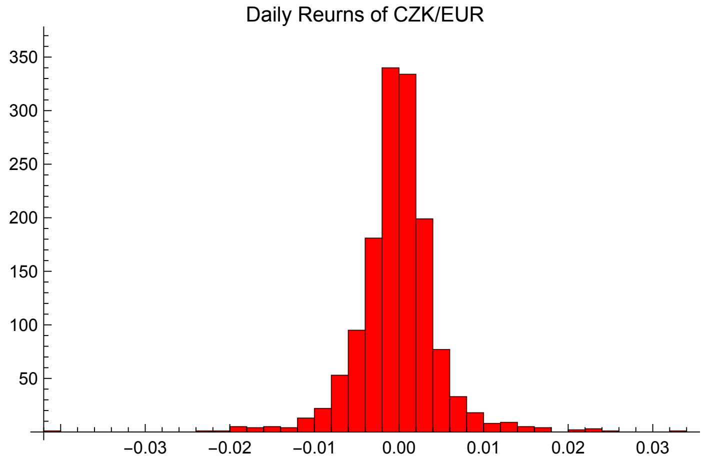


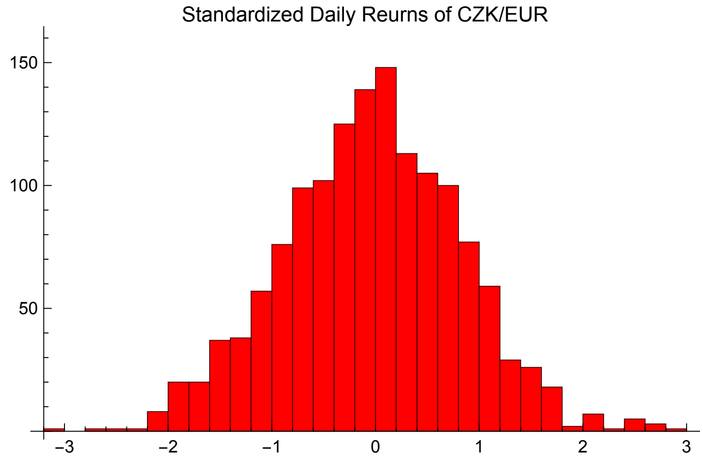






Now Let;s have a look on the distribution of standardized returns $\frac{r_t}{\sqrt{RV_t}}$





Problems

In practice, you can see lot's of problems:
- jumps (you could see it from previous example)
- noise (what is proper sampling? 1min? 5min?...)

RV becomes consistent when $n \longrightarrow 0$, hence $m \longrightarrow \infty$
But we can not sample continuously $\Longrightarrow$ bias in the estimator

Continuous-time no-arbitrage price processes

Let's formalize the previous debate. Most influential work is:
Andersen, Bollerslev, Diebold, Labys: "Modeling and Forecasting Realized Volatility" Econometrica, 2003

Generally, we consider univariate log-price process p_t on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ evolving continuously over time interval [0,T]. $\{\mathcal{F}_t\}$ will be σ-field reflecting information at time t.

Return Decomposition
Any arbitrage-free logarithmic price process p_t represented by a special semi-martingale process can be uniquely represented by:

$r_t = p_t - p_0 = \mu_t + M_t = \mu_t + M_t^c + M_t^J$,
where μ_t is predictable component, M_t is local martingale that can be further decomposed to M_t^c, continuous martingale and M_t^J, jump martingale.

Quadratic Variation of simple Black-Scholes model

Consider a process, where μ is constant, the continuous martingale M_t^c is standard Brownian motion process, and the jump martingale M_t^J, is zero:

$d p_t = \mu\, d t + \sigma\, d W_t,$

Quadratic Variation of this process will be:

$\mathrm{QV}_t = \int_0^t \sigma_s^2\, ds = \sigma^2$
thus, return volatility is constant

Quadratic Variation of Jump-diffusion process

Consider example of jump-diffusion process from last lecture:
$d p_t = \mu_t\, d t + \sigma_t\, d W_t + c_t\, d J_t,$

Quadratic Variation of this process will be:

$\mathrm{QV}_t = \int_0^t \sigma_s^2\, ds + \sum_{s=1}^t J_s^2$
which is so-called *integrated variance* part and variation of jumps.

Asymptotic Result

In case of the $d p_t = \mu_t\, d t + \sigma_t\, d W_t + c_t\, d J_t,$

we have

$\mathrm{RV}_t \longrightarrow \int_0^t \sigma_s^2\, ds + \sum_{s=1}^t J_s^2$

Realized Volatility is unbiased and consistent estimator of Integrated Variance, if $n \longrightarrow 0$, $(m \longrightarrow \infty)$

$\frac{r_t}{\sqrt{\mathrm{RV}_t}} \sim N(0, 1)$

Jump- and Noise-robust volatility estimation

While RV is estimator of total Quadratic Variation,

$\mathrm{QV}_t = \int_0^t \sigma_s^2\, ds + \sum_{s=1}^t J_s^2$

we are mainly interested in separating these two components and understading “continuous”, or integrated volatility, and jumps separately

Jumps - MedRV

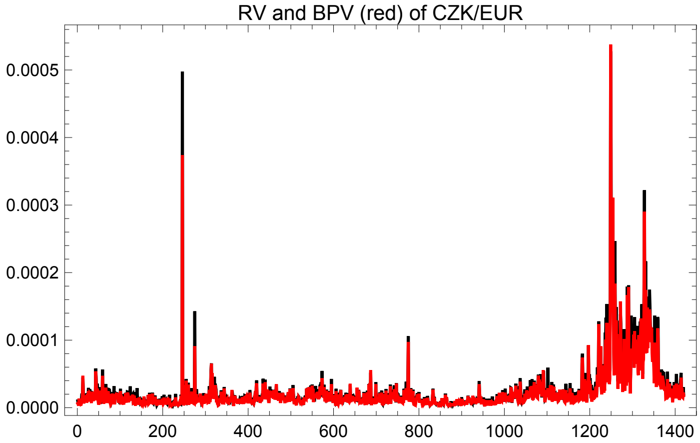
for asset i on day t can be constructed
as:

$\mathrm{MedRV}_{i,t} = \left(\frac{\pi}{6-4\times\sqrt{3}+\pi}\;\frac{M}{M-2}\right) \sum_{j=2}^m \mathrm{med}(|r_{i,t-1+(j-1)\,n}|, |r_{i,t-1+j\,n}|, |r_{i,t-1+(j+1)\,n}|)^2$
for $t=1,\ldots,T$days.

Jumps - Bipower Variation

for asset i on day t can be constructed as:

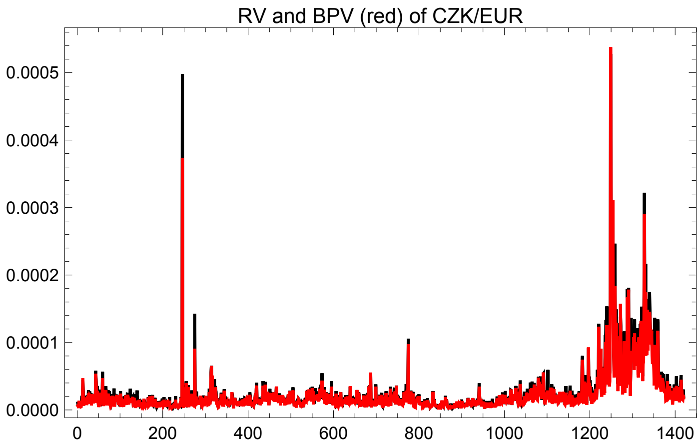
$BPV_{i,t} = \pi/2 \sum_{j=2}^{m} |r_{i,t-1+(j-1)\,n}|.|r_{i,t-1+j\,n}|$
for $t=1,...,T$days.



Noise correction: Bipower Variation

for asset i on day t can be constructed as:

$BPV_{i,t} = \frac{m/(m-2)}{(2/\pi)} \sum_{j=3}^{m} |r_{i,t-1+(j-2)\,n}|.|r_{i,t-1+j\,n}|$
for $t=1,...,T$days.



Testing presence of Jumps

Finally, as RV estimates total quadratic variation, BPV or MedRV integrated volatility IV, their difference are jumps:

$\text{plim}_{m\to\infty} (RV_{i,t} - IV_{i,t}) = JV_{i,t}$

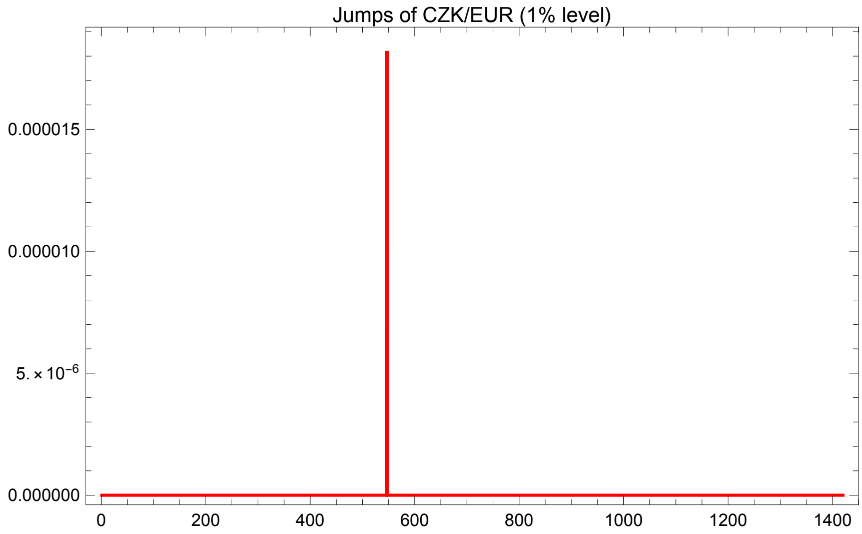
Under the assumption of no jumps and some mild regularity conditions, joint asymptotic distribution of the jump variation is

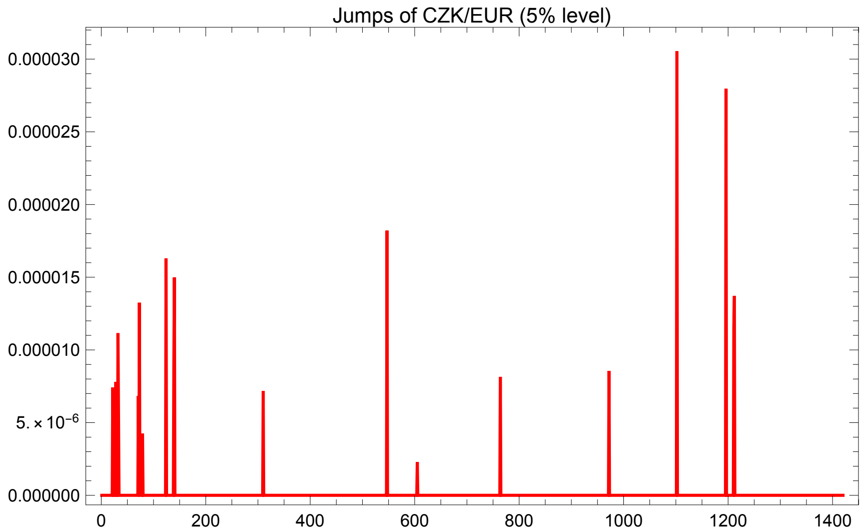
$JV_{i,t} = I_{Z_{i,t} > \Phi(\alpha)} (RV_{i,t} - IV_{i,t})$
where $\Phi(\alpha)$ is critical value from the N(0,1) distribution, and $Z_{i,t}$ is statistics used for testing jumps under the null hypothesis of no jumps (note that it is simple Hausman-type test):

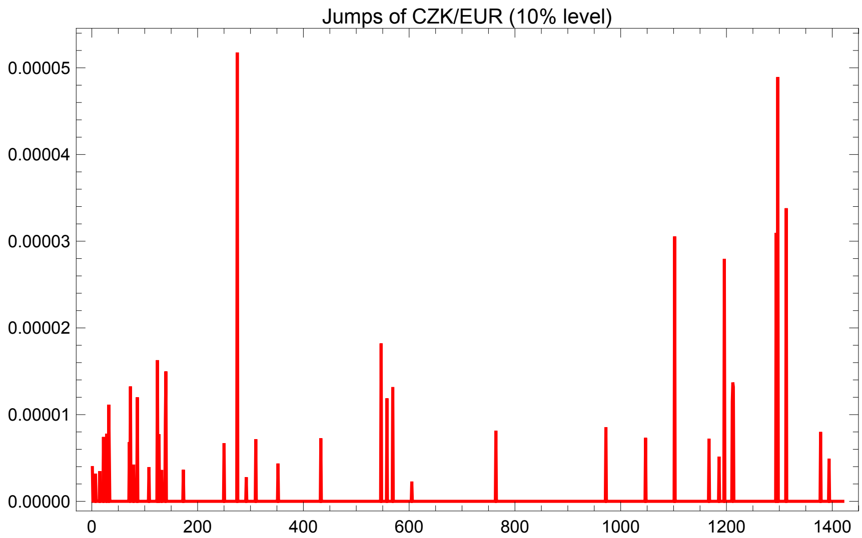
$Z_{i,t} = \frac{\mathrm{RV}_{i,t} - \mathrm{IV}_{i,t}}{\sqrt{(\theta-2) * 1/N * \mathrm{IQ}}}$

where $\mathrm{IQ}_{i,t}$ is estimate of fourth moment for example as

$\mathrm{TQ}_{i,t} = \frac{m^2/(m-4)}{(0.8313)^3} \sum_{j=5}^{m} |r_{i,t-1+(j-4)\,n}|^{4/3} . |r_{i,t-1+(j-3)\,n}|^{4/3} . |r_{i,t-1+(j-2)\,n}|^{4/3}$


Jumps of CZK/EUR (1% level)
0.000015
0.000010
5. × 10^-6
0.000000
0
200
400
600
800
1000
1200
1400


Jumps of CZK/EUR (5% level)
0.000030
0.000025
0.000020
0.000015
0.000010
5. × 10^-6
0.000000
0
200
400
600
800
1000
1200
1400


Jumps of CZK/EUR (10% level)
0.00005
0.00004
0.00003
0.00002
0.00001
0.00000
0
200
400
600
800
1000
1200
1400

Noise: subsampling

One can also estimate RV with different frequencies and average. For example one can consider δ>5 interval and δ/5 subsamples of δ-minute returs. We can have three 5-minute intervals within 15 minutes producing three subsamples

Realized Covariance & Correlation

Let's consider generalization realized covariance estimation of the two assets *a* and *b*:

$\text{RCov}_t = \sum_{j=1}^{m} r_{a,t-1+j\,n}\, r_{b,t-1+j\,n}$
for $t=1,\ldots,T$days.

Again, with $n \longrightarrow 0$, $(m \longrightarrow \infty)$, this estimator will converge in probability to the covariance matrix of two assets *a* and *b*: $\int_0^1 \sum_{a,b} d s$

Realized Correlation can then be estimated as:

$\text{RC}_t = \frac{\text{RCov}_{t,a,b}}{\sqrt{\text{RV}_{t,a}}\,\sqrt{\text{RV}_{t,b}}}$